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ADDING VALUE TO ENGINEERING



Skill Gap Analyzer

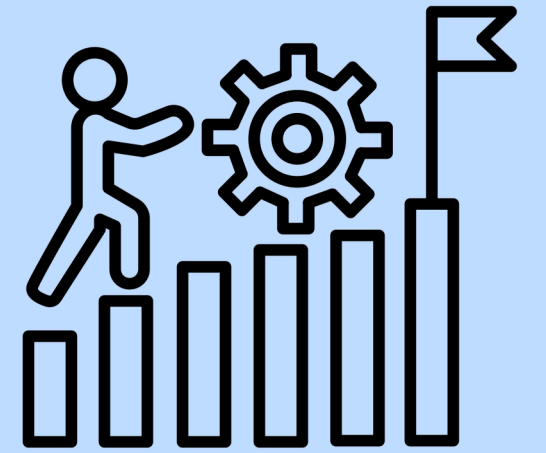
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PROBLEM STATEMENT

In today's **competitive job market**, students and professionals lack a quick, automated tool to identify precisely which skills they need to acquire in order to qualify for a desired role.

Manually comparing skill sets against job descriptions is error-prone and time-consuming.



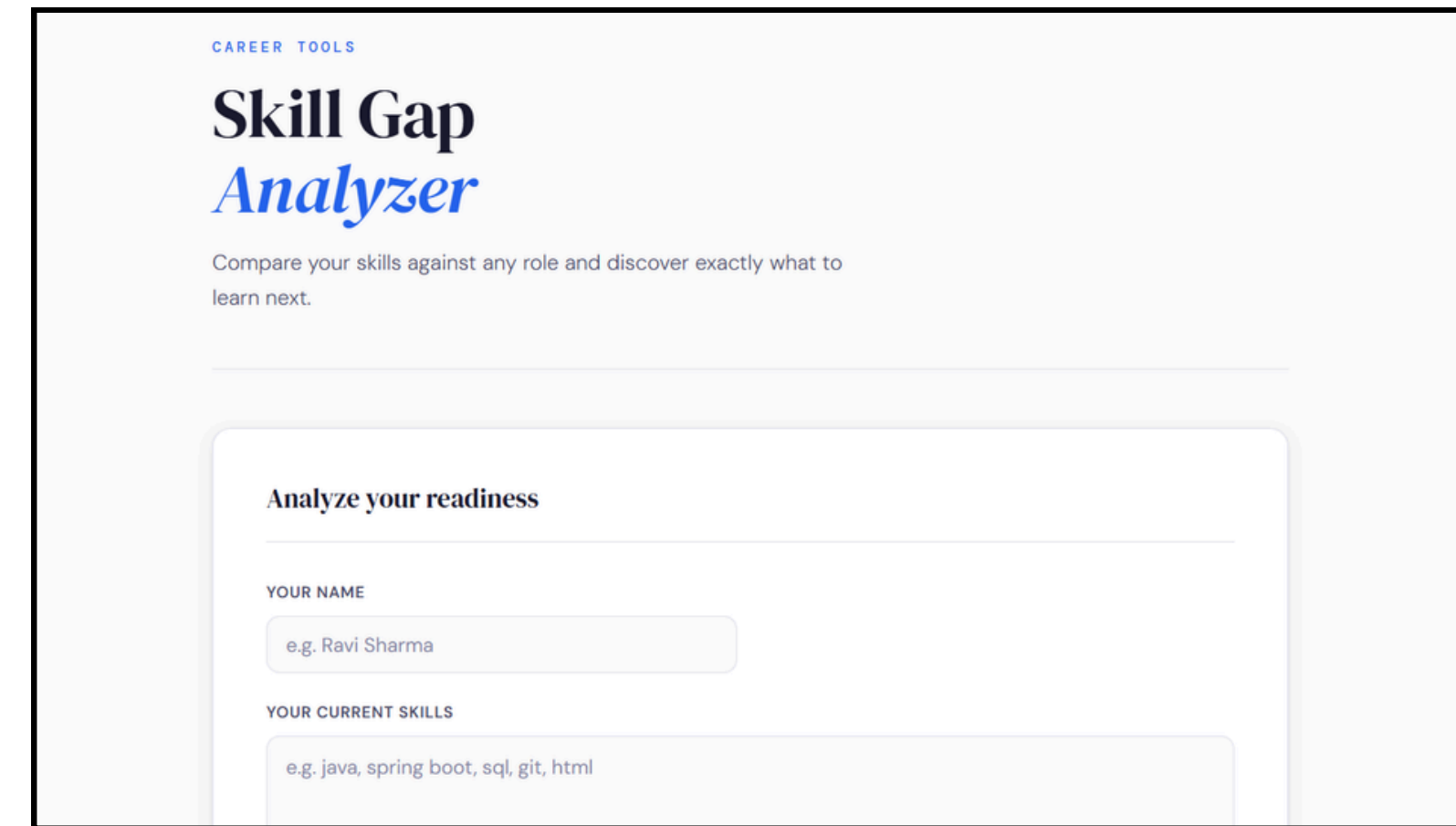
This project solves the problem by:

- Accepting the user's current skill set as input.
- Fetching required skills for a target role from a JSON-based simulated API (roles.json).
- Comparing the two sets and producing a detailed gap report.
- Calculating a readiness percentage score with a visual progress bar.
- Providing a browser-based frontend (HTML + CSS + JS) for interactive analysis.

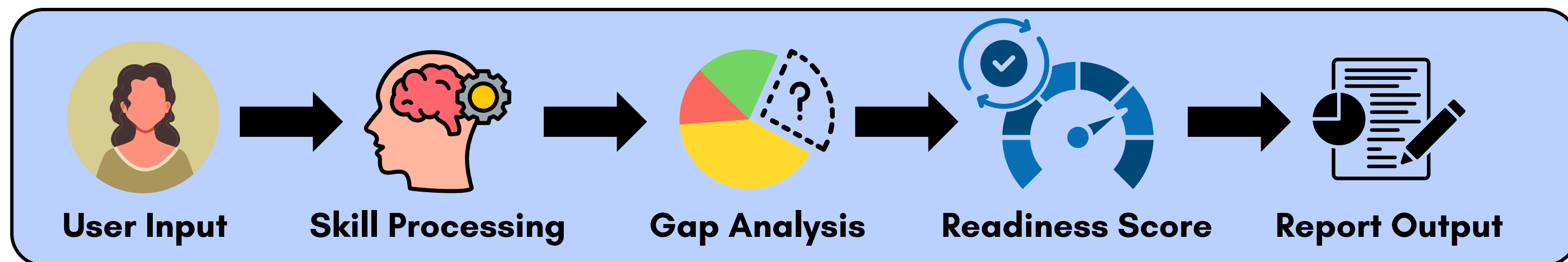
SOLUTION

A full-stack application called “**Skill Gap Analyzer**” that:

- Takes user's current skills as input.
- Compares them against required skills for a target job role.
- Reports missing skills with a readiness percentage score.
- Provides interactive browser-based frontend.
- Built using core Java OOP principles, Maven, JSON-based API simulation, and HTML,CSS and JS.



The screenshot shows the 'Skill Gap Analyzer' web application. At the top, it says 'CAREER TOOLS' and 'Skill Gap Analyzer'. Below the title, it says 'Compare your skills against any role and discover exactly what to learn next.' The main form area is titled 'Analyze your readiness' and contains two input fields: 'YOUR NAME' with a placeholder 'e.g. Ravi Sharma' and 'YOUR CURRENT SKILLS' with a placeholder 'e.g. java, spring boot, sql, git, html'.



OOP CONCEPTS USED

1. Abstraction

- ApiService is an abstract class
- Declares fetchRoles() without implementing it
- SkillGapAnalyzer never knows if data comes from a file

4. Polymorphism

SkillGapAnalyzer holds an ApiService reference
At runtime, JsonApiService is injected dynamically
Swap to RealApiService in future with just one line change

2. Encapsulation

All fields in Skill, User, Role are private
Accessible only via public getters/setters
User.addSkill() enforces no-duplicate rule internally

5. Method Overriding

- Skill.equals() overridden for case-insensitive matching
- "Java", "java", "JAVA" all treated as the same skill

3. Inheritance

JsonApiService extends ApiService
Inherits sourceName field and fetchRoleByName() method
Overrides the abstract fetchRoles() to read from JSON

6. Collections

- ArrayList<Skill> used in User and Role
- .contains() triggers Skill.equals() for smart comparison

TECHNOLOGY USED

Layer	Technology
Core Language	Java 11+
Build Tool	Apache Maven 3.8+
Data / API Simulation	JSON (roles.json)
Frontend	HTML5 + CSS3 + Vanilla JavaScript
IDE	IntelliJ IDEA / Eclipse / VS Code
JSON Library	org.json (via Maven)
OS Support	Windows 10/11, macOS 12+, Ubuntu 20.04+

FEATURES AND USP

- Readiness Score — calculates % match with animated progress bar
- Custom Role Builder — define your own target role & skills
- Smart Skill Matching — case-insensitive comparison using Skill.equals()
- Dual Interface — works on both Java console & browser
- Difficulty Badges — skills tagged Beginner / Intermediate / Advanced
- Extensible Design — add roles via JSON, no core logic changes
- Offline Frontend — browser app runs with zero server needed

REFERENCES / LINKS

2. **PRD** :

10. **FFmpeg** - <https://ffmpeg.org/>

11. **Google Colab** - <https://colab.research.google.com/>

12. **ngrok** - <https://ngrok.com/>

Github Link: <https://github.com/osinsomkuwar-27/aiavatar>

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Thank You

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